

UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Scienze Fisiche,
Informatiche e Matematiche

HPC Labs - Introduction

High-Performance Computing [262-022]

Corso di Laurea in INFORMATICA
(D.M.270/04) [16-262]
Anno accademico 2026/2027

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Dr. Alessandro Capotondi



RTD-A (Assistant Professor)

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https://unimore.unifind.cineca.it/get/person/108674

Office: MO-18-03-009 (third floor)
PhD from **University of Bologna**
PostDoc from **University of Modena**
Visiting at **ETH Zurich** (CH)

Research interests: heterogeneous embedded systems, programming models, hardware acceleration, embedded machine learning

Student reception: By appointment, to be agreed via email

Thesis on:

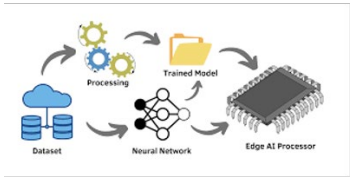
- TinyML and edge AI
- Continual Learning and On-device training
- LLM, SLM, VLA on the edge
- HW/SW Co-design of Computing Platforms for Autonomous systems

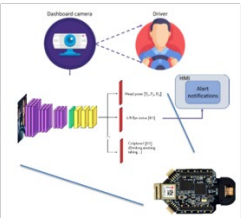
International Internships (duration ~6 months – arrangement in advance is required):

- KU Leuven
- ETH Zurich
- SUPSI Lugano
- INRIA Lione











System Integration
And Applications

Programming Models

NVIDIA CUDA

OpenCL

OpenMP

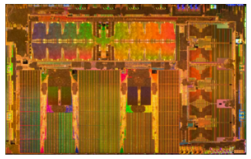
NVIDIA

GCC

Compiler Extensions

LLVM
COMPILER INFRASTRUCTURE

PULP
Parallel Ultra Low Power



RISC-V
Hardware Architecture
Extensions and
Exploration

AMD
XILINX

AXELERA
ARTIFICIAL INTELLIGENCE

ETH zürich

Università della Svizzera italiana

Scuola universitaria professionale della Svizzera italiana
SUPSI

KU LEUVEN

Objectives

- Learn how to improve application performances exploiting modern parallel architectures.

- Learn to use different parallel programming models
  - OpenMP (basic to intermediate)
  - CUDA (basic to intermediate)
  - HLS (basic)

- Learn to evaluate, design, implement and present accelerated application on real (near-to) use-cases

NVIDIA Jetson Developer Kits

Easy to Develop, Faster to Market



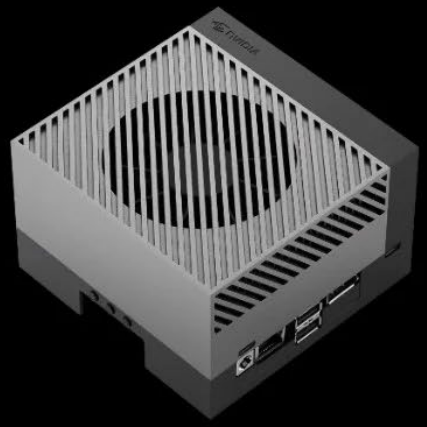
Jetson Nano Developer Kit

0.472 TFLOPS
$149



Jetson Orin Nano Developer Kit

40 TOPS (INT8)
$499



Jetson AGX Orin Developer Kit

275 TOPS (INT8)
$1999

LEARNER | INTERMEDIATE | ADVANCED

		
Development Kit	NVIDIA Jetson Nano B01 Dev Kit	NVIDIA Jetson Orin Nano 8GB Dev Kit
AI performance	472 GFLOPS	40 TOPS
GPU	NVIDIA Maxwell™ Architecture 128 x core	NVIDIA Ampere™ Architecture 1024 x NVIDIA CUDA core 32 x 3rd Gen Tensor core
CPU	Quad-core ARM Cortex-A57 @ 1.43 GHz	6-core ARM Cortex-A78AE v8.2 64-bit @ 1.5 GHz (max)
Memory	4GB 64-bit LPDDR4 @ 25.6GB/s	8GB 128-bit LPDDR5 @ 68GB/s

Lab Configuration

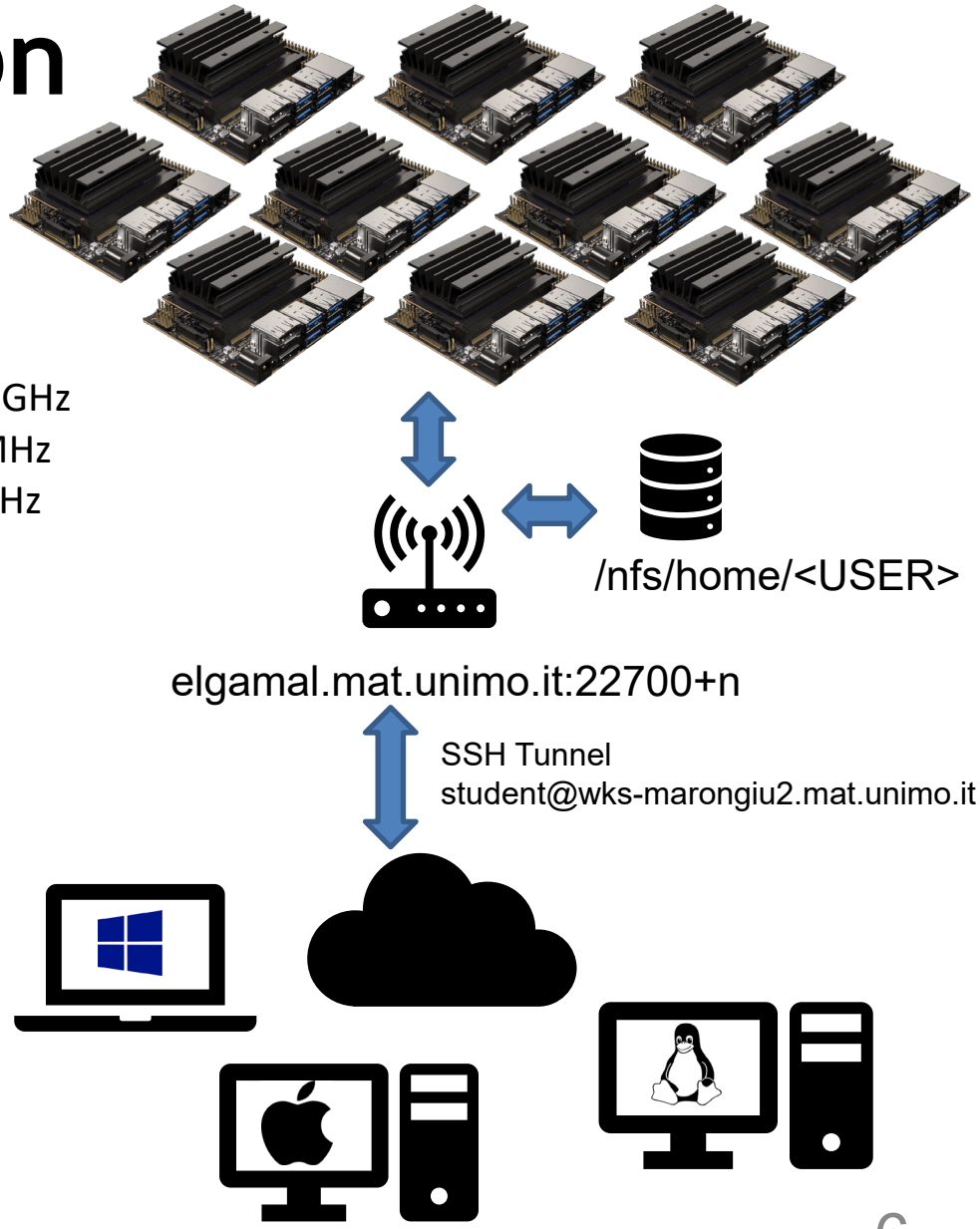
8x Nvidia Jetson Nano

Hardware

CPU 64-bit Quad-core ARM A57 @1.43GHz
GPU 128-core NVIDIA Maxwell @921MHz
Memory 4GB 64-bit LPDDR4 @ 1600MHz
32/16 GB Local Storage

Software

Ubuntu 18.04 LTS OS
Nvidia Jetpack 4.3
Cuda 10.0
NFS User Home



SSH Tunnel

Tunnel: student@wks-marongiu2.mat.unimo.it
Port: 22

URL: elgamal.mat.unimo.it
Port: 22700+<BoardID>

List of Boards

Asset	URL	Port	Guest	Passw	Note
nano0	elgamal.mat.unimo.it	22700	student	hpc2026	Ubuntu 20.04 LTS - 32GB
nano1	elgamal.mat.unimo.it	22701	student	hpc2026	Ubuntu 20.04 LTS - 32GB
nano2	elgamal.mat.unimo.it	22702	student	hpc2026	Ubuntu 20.04 LTS - 16GB
nano3	elgamal.mat.unimo.it	22703	student	hpc2026	Ubuntu 20.04 LTS - 32GB
nano4	elgamal.mat.unimo.it	22704	student	hpc2026	Ubuntu 20.04 LTS - 32GB
~~**nano5**~~	~~elgamal.mat.unimo.it~~	~~22705~~	~~student~~	~~hpc2026~~	~~Ubuntu 18.04 LTS - 32GB~~
nano6	elgamal.mat.unimo.it	22706	student	hpc2026	Ubuntu 20.04 LTS - 16GB
nano7	elgamal.mat.unimo.it	22707	student	hpc2026	Ubuntu 20.04 LTS - 32GB
nano8	elgamal.mat.unimo.it	22708	student	hpc2026	Ubuntu 20.04 LTS - 32GB
nano9	elgamal.mat.unimo.it	22709	student	hpc2026	Ubuntu 20.04 LTS - 32GB
~~**orin**~~	~~172.19.0.61~~	~~22~~	~~student~~	~~hpc2026~~	~~Ubuntu 22.04 LTS - 500GB~~
~~**orin1**~~	~~172.19.0.82~~	~~22~~	~~student~~	~~hpc2026~~	~~Ubuntu 22.04 LTS - 500GB~~

Software Configuration

- **Jetpack L4T R32.7.6**
  - CUDA 10.0
  - TensorRT 6
  - cuDNN 7.6
  - OpenCV 4.1
- **CUDA Environment setup**

```
module load cuda/10.0
```

- **CLANG/LLVM with OpenMP 4.5 Support (GPU-offloading)**

```
module load clang/11.0.0
```

How-to Connect to a board – Your USERNAME

- **Using your personal account**
  - **Every student** registered in Moodle has a personal account.
  - The **home is shared** between the different boards. You will find the same files on every nano you login.
  - The account **<username>** is the one of your UNIMORE email (i.e. 1234567@studenti.unimore.it, the **<username>** is 1234567)
  - The **password** is your student ID (i.e. {1}7654321, then your password is 7654321)
  - For users with @**UNIMORE.IT** account (e.g., PhD Students) the password is **hpc2026**

- **Using a guest account**
  - Username: **student**
  - Password: **hpc2025**
  - **NOTE: guest account is not a NFS shared folder.**

ACTION REQUIRED

Register your presence and your SSH public key!

https://forms.gle/86ee5SUoxDGEy7f87

Board Schedule

During Lectures

- **Students** can access the boards according to the following allocation: LINK

- **In any other period**
  - **ALL the students** can access any boards freerly



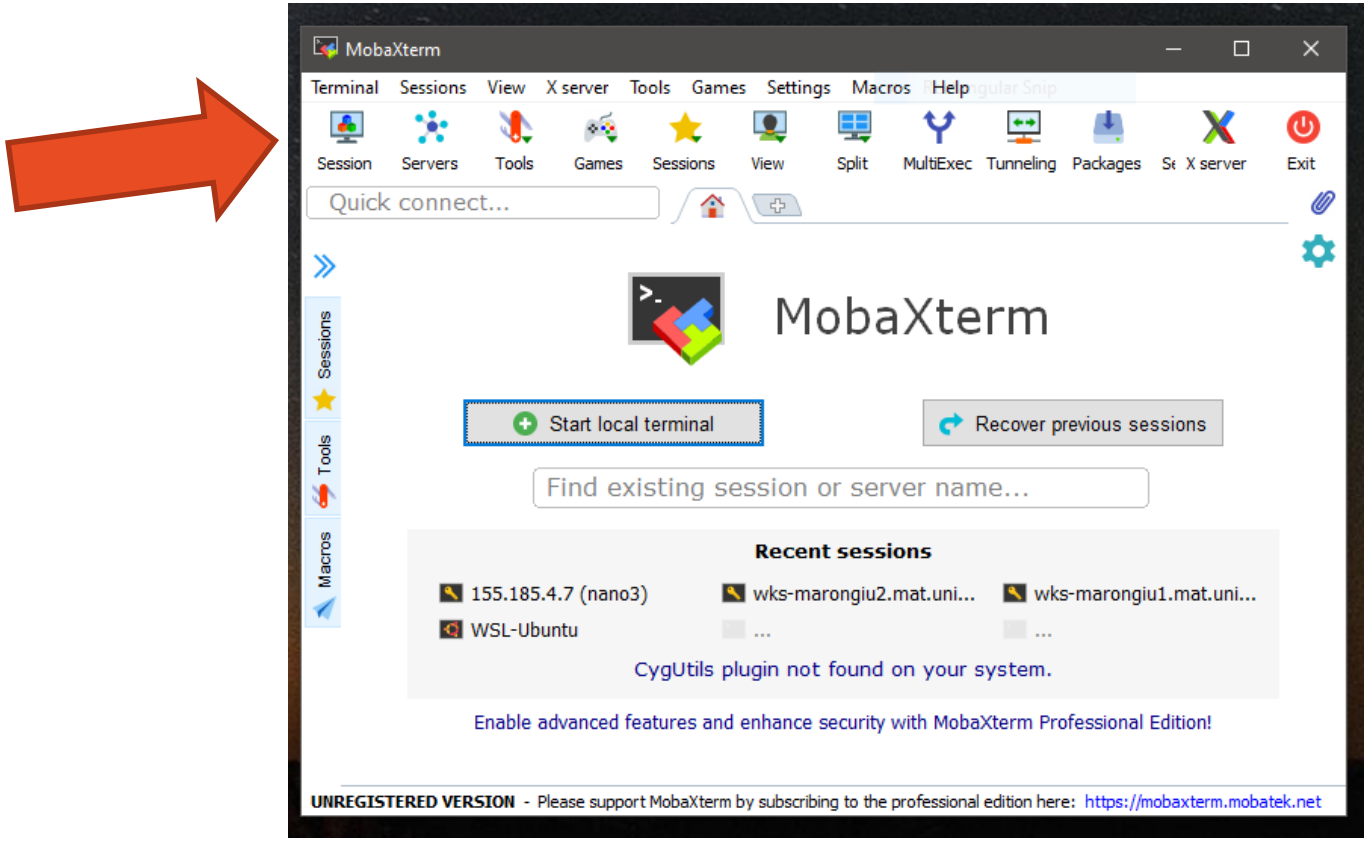
https://www.youtube.com/watch?v=bNCNp0khR6M

Your software requirements

- **Windows-Users**
  - MabaXterm – Supported (https://mobaxterm.mobatek.net/download.html)
  - Putty (https://www.putty.org/)
  - Windows Ubuntu Bash (https://ubuntu.com/tutorials/tutorial-ubuntu-on-windows#1-overview)
- **Mac-Users**
  - Native SSH Client
  - https://osxdaily.com/2017/04/28/howto-ssh-client-mac/
- **Linux-Users:**
  - Native SSH Client

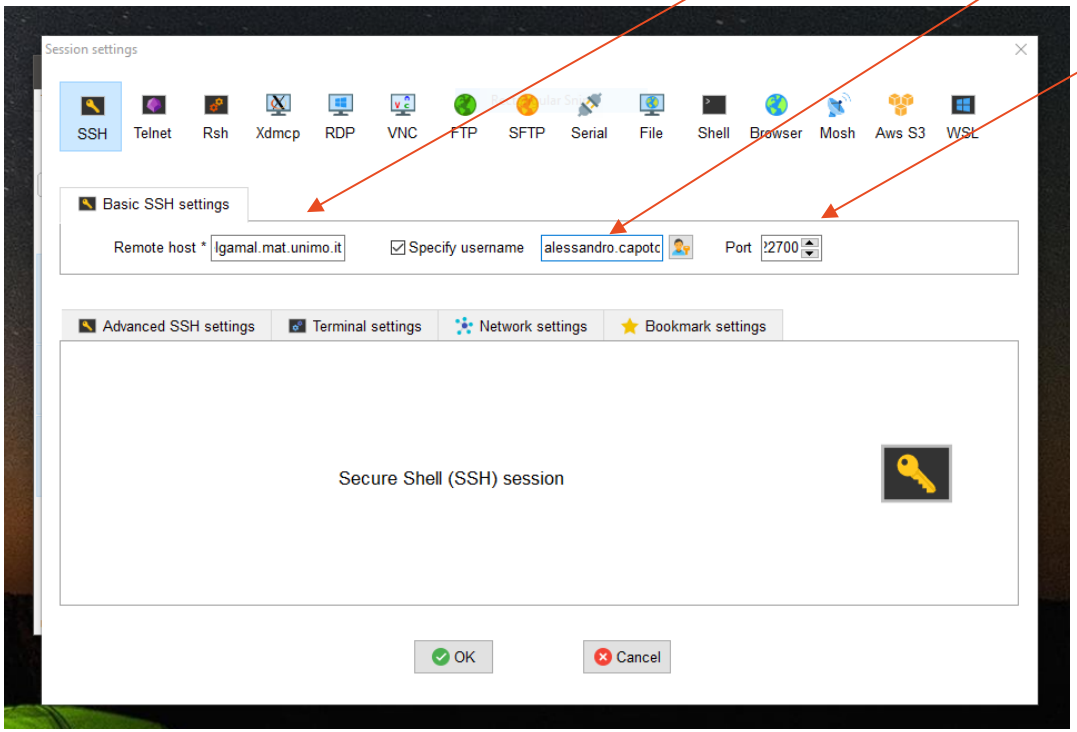
How-to Connect to a board - Windows

1. **Create New Session**



How-to Connect to a board - Windows

1. **Open MobaXterm**
2. **Create New SSH Session**
3. **Add Your Configuration**



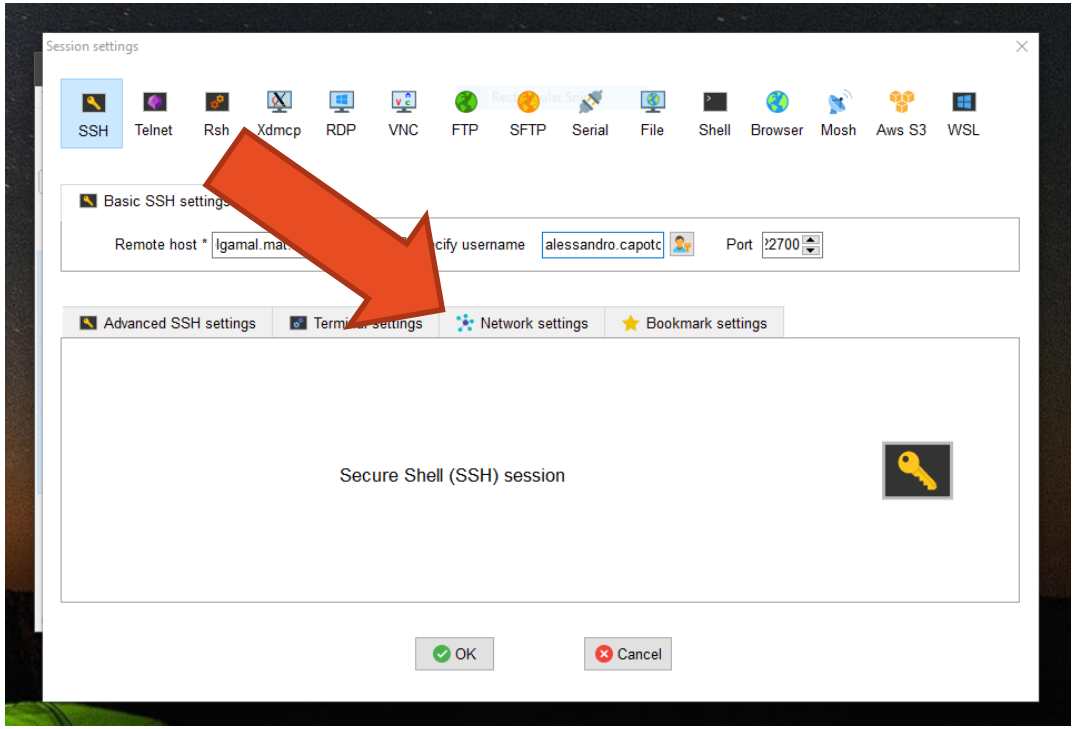
elgamal.mat.unimo.it

<USERNAME>
(e.g alessandro.capotondi)

<BOARD PORT>
(e.g. 2270N (N: 0-9))

How-to Connect to a board - Windows

1. **Open MobaXterm**
2. **Create New SSH Session**
3. **Add Your Configuration**
4. **Add Proxyjump Configuration (Under Network Settings)**

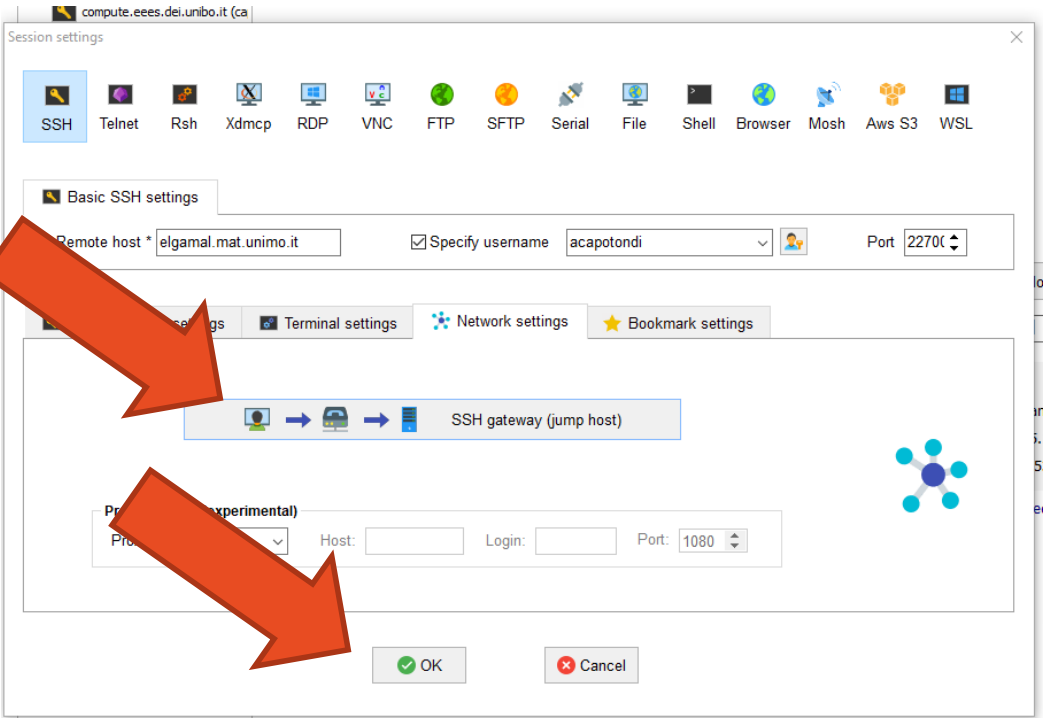


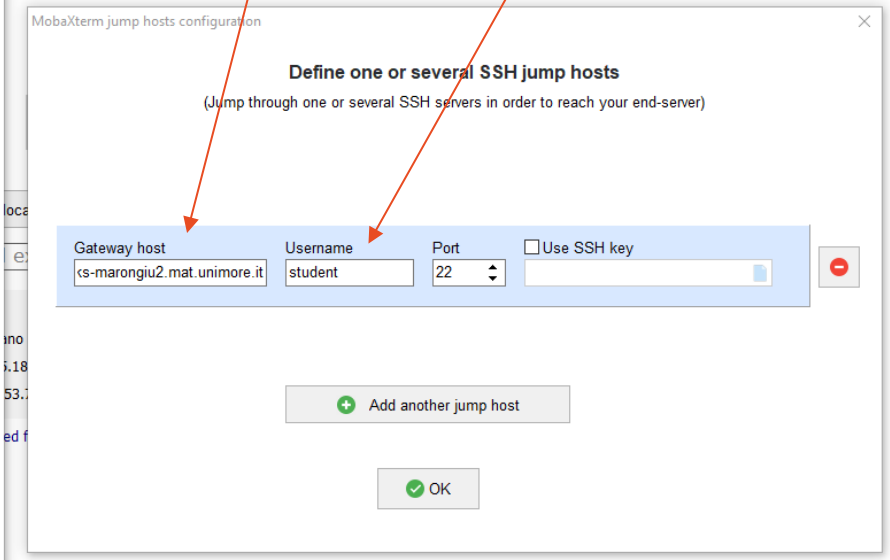
How-to Connect to a board - Windows

1. **Open MobaXterm**
2. **Create New SSH Session**
3. **Add Your Configuration**
4. **Add Proxyjump Configuration (Under Network Settings)**

wks-marongiu2.mat.unimore.it

student





How-to Connect to a board - Windows

1. **Open MobaXterm**
2. **Create New SSH Session**
3. **Add Your Configuration**
4. **Add Proxyjump Configuration (Under Network Settings)**
5. **Prompt Password for the Proxyjump: *hpc2026***
6. **Prompt Your User Password**

How-to Connect to a board – Linux/Mac

1. Open Terminal
2. Create the SSH Tunnel

```
ssh -N -L <PORT>:elgamal.mat.unimo.it:<PORT> student@wks-marongiu2.mat.unimo.it -v
```

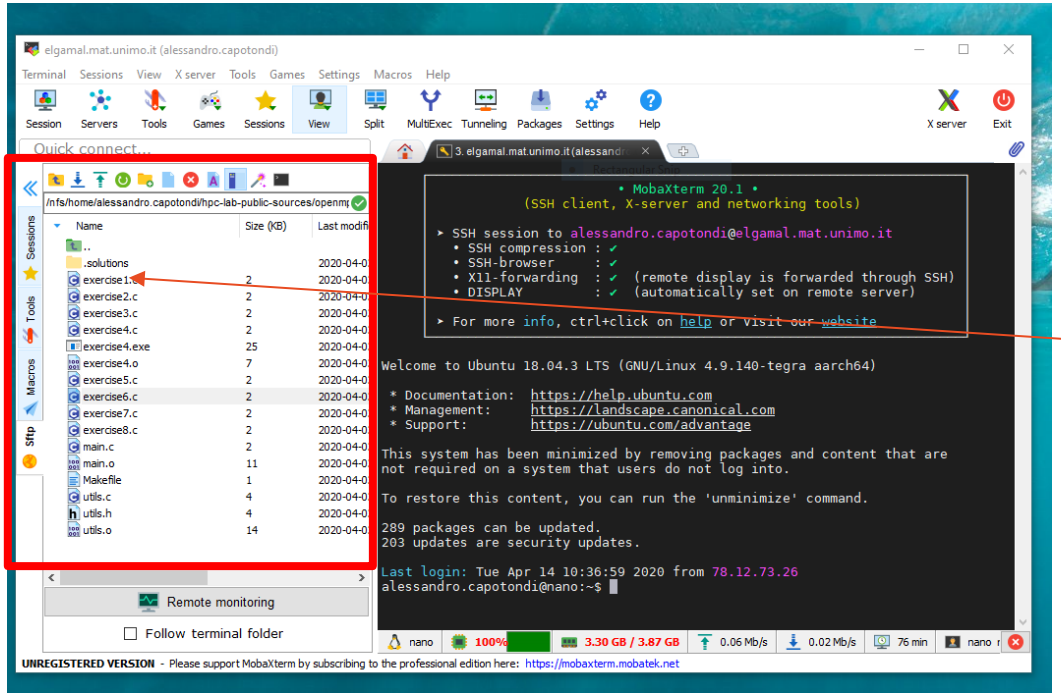
- ➔ *SSH Tunnel URL:* *wks-marongiu2.mat.unimo.it*
- ➔ *User:* *student*
- ➔ *Password:* *hpc2026*

3. Open a new Terminal
4. Connect through SSH client

```
ssh -X -p <PORT> <USERNAME>@localhost
```

Lab Improve Productivity

- Mount your board home in your local machine using SSH:
  - **GNU/Linux – MacOS Users:** https://www.digitalocean.com/community/tutorials/how-to-use-sshfs-to-mount-remote-file-systems-over-ssh
  - **Windows Users:** you can use Mobaxterm's SFTP built-in client.



Right-Click on the file:

– Open with…

select your local preferred editor



Register your presence and your SSH public key!

https://forms.gle/86ee5SUoxDGEy7f87

Download, Compile and Run

1. Connect to your assigned Nano board: (See HPC-LAB-HOWTO)

```
ssh -X -p PORT <USER>@<HOST>
```

2. Download Sources

```
git clone https://git.hipert.unimore.it/hpc_class/hpc-lab-public-sources.git
```

3. Compile and Run

```
cd hpc-lab-public-sources/openmp/lab1
make EXERCISE=exercise1.c EXT_CFLAGS="-DNTHREADS=4" clean all run
```

Select the exercise source

You can pass custom compilation flags

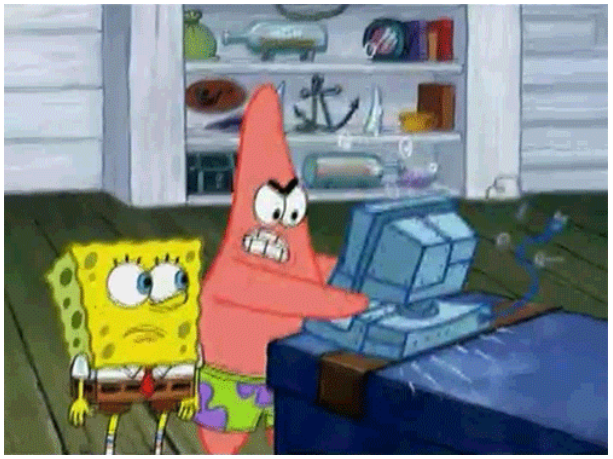
```
nano3@nano3:~$ tree hpc_lab/
hpc_lab/
├── LICENSE
├── openmp
│   └── lab1
│       ├── exercise1.c
│       ├── exercise2.c
│       ├── exercise3.c
│       ├── exercise4.c
│       ├── exercise5.c
│       ├── exercise6.c
│       ├── main.c
│       ├── Makefile
│       ├── utils.c
│       └── utils.h
└── README.md
```

Support and Assistance

If you have problem to login, or with your account, or any other related to the lab.

Sent an email with the object “[HPC2026] Remote Lab Assistance” to a.capotondi@unimore.it and andrea.marongiu@unimore.it or click this link.

We will fix it…



Appendix

Lab @HOME

HW Requirements

- Nvidia GP-GPU
- GNU/Linux OS

SW Requirements

- Nvidia Drivers Installation
- Nvidia CUDA Tools Installation
- OpenMP4.5 or OpenMP5 Compiler ***(Compiler MUST be rebuilt)***
  - **(suggested)** clang/LLVM
    - https://hpc-wiki.info/hpc/Building_LLVM/Clang_with_OpenMP_Offloading_to_NVIDIA_GPUs
  - GNU GCC
    - https://gist.github.com/matthiasdiener/e318e7ed8815872e9d29feb3b9c8413f

Remote Nsight Usage (ubuntu-18.04)

a) On your workstation: install CUDA Toolkit 10.0 (same version of the Jetson Nano): https://developer.nvidia.com/cuda-10.0-download-archive

b) On your workstation: install Cross-Compilation Tools:

```
sudo dpkg --add-architecture arm64
sudo apt-get update
sudo apt-get install cuda-cross-aarch64 g++-aarch64-linux-gnu
gcc-aarch64-linux-gnu openjdk-8-jre
```

For other OS:
https://docs.nvidia.com/cuda/nsight-eclipse-edition-getting-started-guide/index.html#using-nsight

Nsight Launch

- **Launch Nsight on your workstation**:

```
nsight -vm /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java
```

- Workflow:

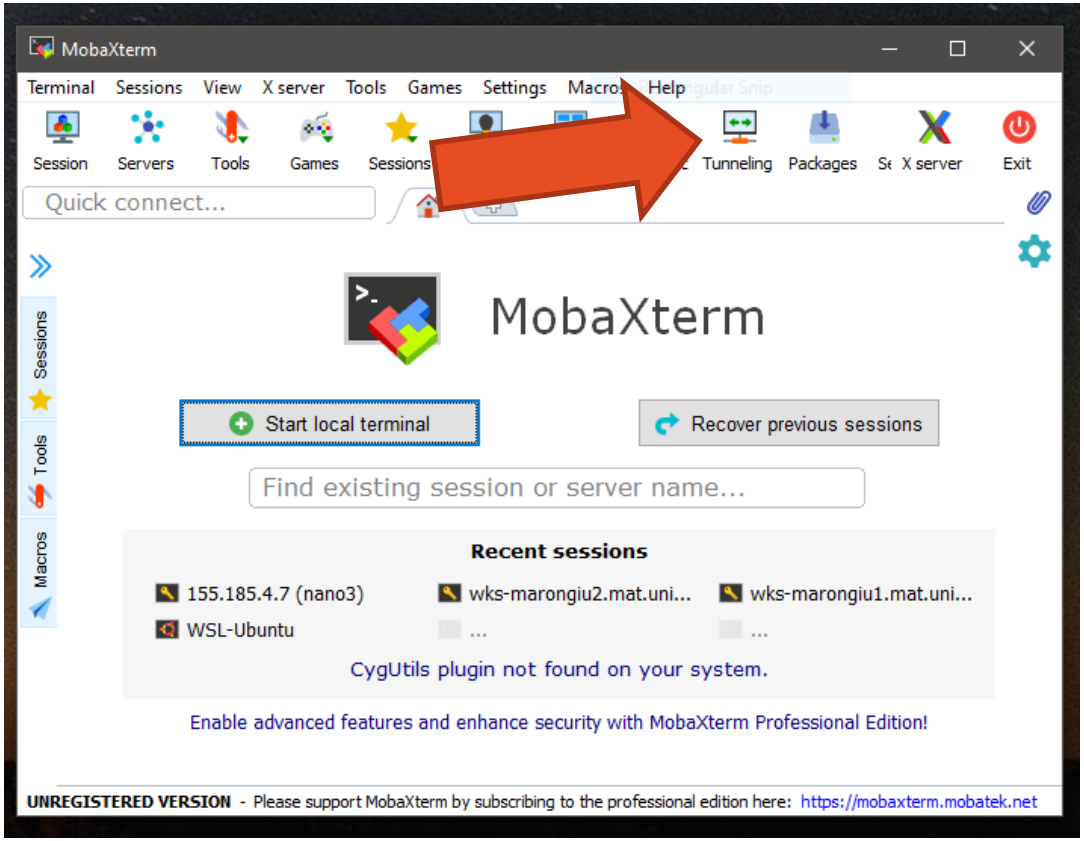


Look the video for the remote connection (in Italian)
https://eu.bbcollab.com/collab/ui/session/playback/load/bd72c36cd9254387868f6bb210be1b2a

DEPRECATED

How-to Connect to a board - Windows

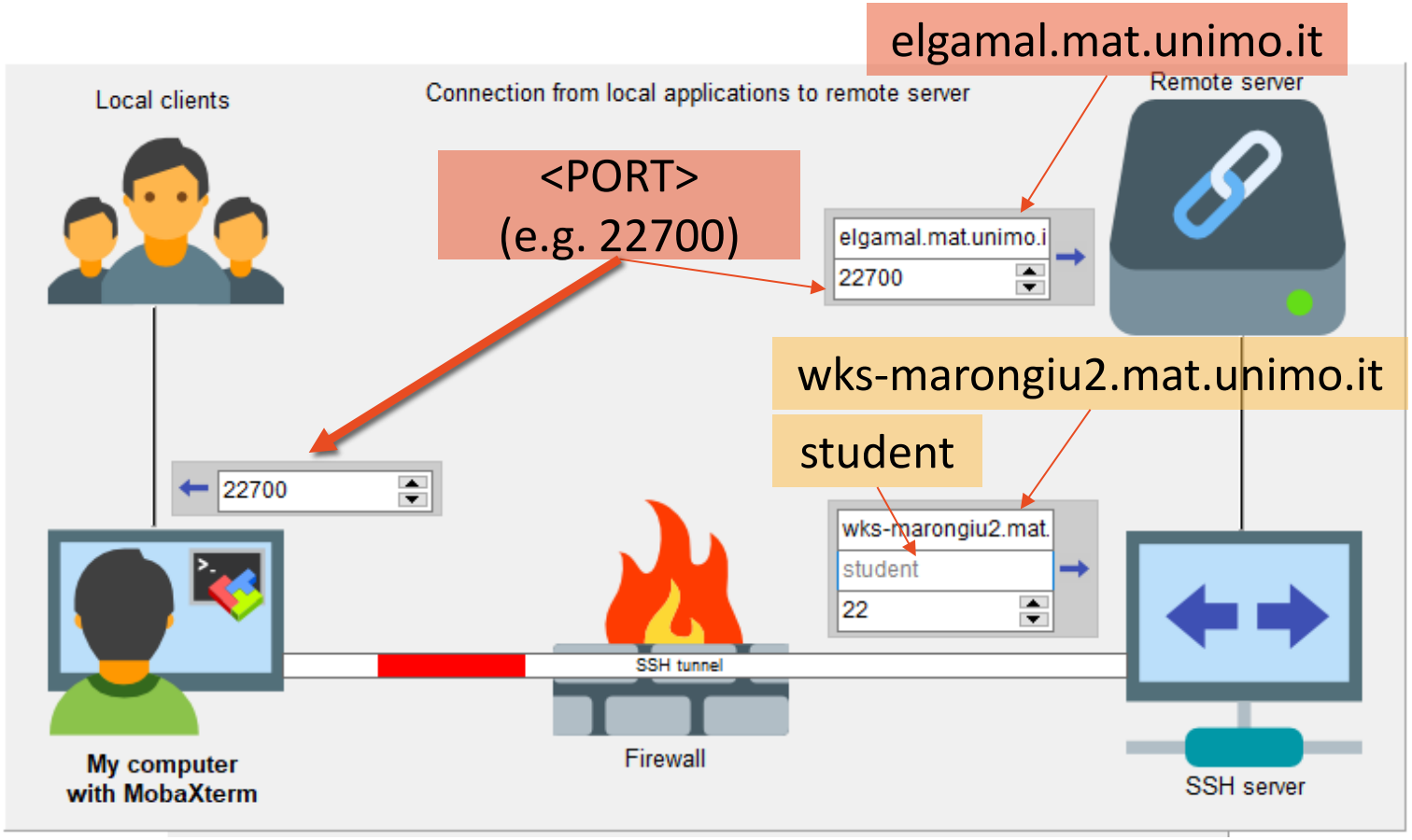
1. **Open MobaXterm**
2. **Configure the SSH Tunnel**



DEPRECATED

How-to Connect to a board – The SSH Tunnel

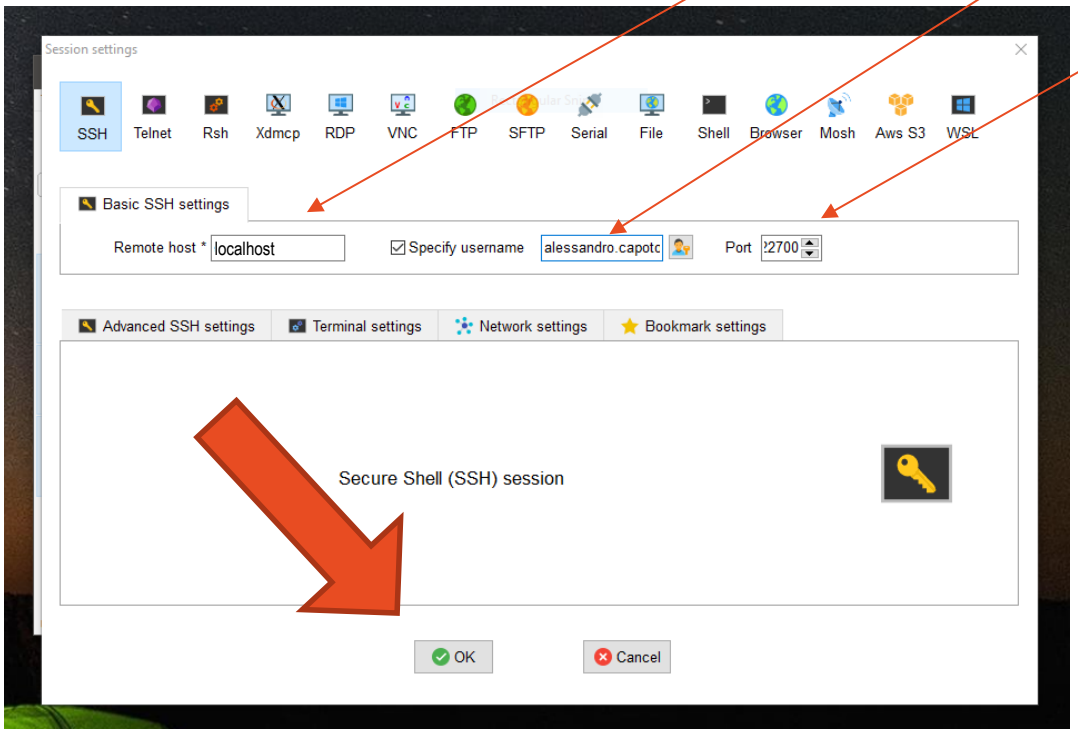
- ➔ *SSH Tunnel URL:* *wks-marongiu2.mat.unimo.it*
- ➔ *User:* *student*
- ➔ *Password:* *hpc2025*



DEPRECATED

How-to Connect to a board - Windows

1. **Open MobaXterm**
2. **Create New SSH Session**
3. **Add Your Configuration**
4. **Prompt Your Password**



localhost

<USERNAME>
(e.g alessandro.capotondi)

<BOARD PORT>
(e.g. 2270N (N: 0-9))